

**A 15-LED CIRCUIT WITH 5 DIFFERENT DISPLAY CONFIGURATIONS BASED
ON BOOLEAN ALGEBRA
DISCRETE MATHEMATICS MIDTERM EXAMINATION**



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Design a circuit consisting of 15 LEDs with 5 different display patterns.

1. The Displays

B	C	D	E
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15
	1	2	3
	4	5	6
	7	8	9
	10	11	12
	13	14	15

2. Their corresponding tables

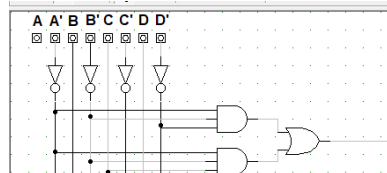
	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Animasi	A	B	C	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
0	0	0	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	1	1	
1	0	0	0	1	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	
2	0	0	1	0	1	1	1	1	0	1	1	1	1	0	0	1	1	1	1	
3	0	0	1	1	1	0	0	1	0	0	1	1	1	1	0	1	1	0	1	
4	0	1	0	0	0	1	0	0	1	0	1	1	1	0	1	0	0	1	1	

3. Karnaugh maps along with the minimized results (<http://www.32x8.com/index.html>)

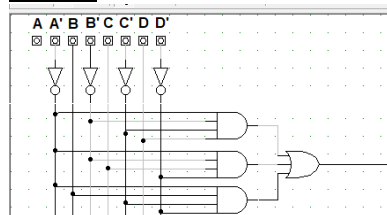
AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR
LED 1 $A'B'D + A'BC$	AB/CD	00	01	11	10	LED 6 $A'B'CD'$	AB/CD	00	01	11	10	LED 11 $A'B'CD + A'BCD'$	AB/CD	00	01	11	10
	00	1	0	1	1		00	0	0	0	1		00	0	1	0	0
	01	0	0	0	0		01	0	0	0	0		01	1	0	0	0
	11	0	0	0	0		11	0	0	0	0		11	0	0	0	0
	10	0	0	0	0		10	0	0	0	0		10	0	0	0	0
LED 2 $A'B'CD + A'BCD' + ABCD$	AB/CD	00	01	11	10	LED 7 $A'CD' + A'BC$	AB/CD	00	01	11	10	LED 12 $A'B'C$	AB/CD	00	01	11	10
	00	0	1	0	1		00	1	0	1	1		00	0	0	0	1
	01	1	0	0	0		01	1	0	0	0		01	0	0	1	0
	11	0	0	0	0		11	0	0	0	0		11	0	0	0	0
	10	0	0	0	0		10	0	0	0	0		10	0	0	0	0
LED 3 $A'B'CD$	AB/CD	00	01	11	10	LED 8 $A'BD + A'BC + ABCD'$	AB/CD	00	01	11	10	LED 13 $A'BD + A'BC$	AB/CD	00	01	11	10
	00	0	0	0	1		00	0	1	1	1		00	1	0	1	1
	01	0	0	0	0		01	1	0	0	0		01	0	0	0	0
	11	0	0	0	0		11	0	0	0	0		11	0	0	0	0
	10	0	0	0	0		10	0	0	0	0		10	0	0	0	0
LED 4 $A'BD + A'BC$	AB/CD	00	01	11	10	LED 9 $A'BC + ABCD'$	AB/CD	00	01	11	10	LED 14 $A'BC + A'BD + A'CD'$	AB/CD	00	01	11	10
	00	1	0	1	1		00	0	0	1	1		00	1	1	0	1
	01	0	0	0	0		01	1	0	0	0		01	1	0	0	0
	11	0	0	0	0		11	0	0	0	0		11	0	0	0	0
	10	0	0	0	0		10	0	0	0	0		10	0	0	0	0
LED 5 $A'B'CD + ABCD'$	AB/CD	00	01	11	10	LED 10 $A'B'CD + A'BCD$	AB/CD	00	01	11	10	LED 15 $A'CD' + A'BC$	AB/CD	00	01	11	10
	00	0	1	0	0		00	1	0	1	0		00	1	0	1	1
	01	1	0	0	0		01	0	0	0	0		01	1	0	0	0
	11	0	0	0	0		11	0	0	0	0		11	0	0	0	0
	10	0	0	0	0		10	0	0	0	0		10	0	0	0	0

4. Screenshots of each display from Digital Works

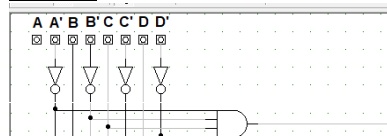
LED 1



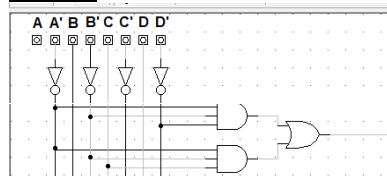
LED 2



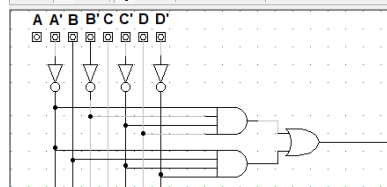
LED 3



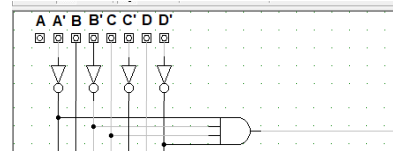
LED 4



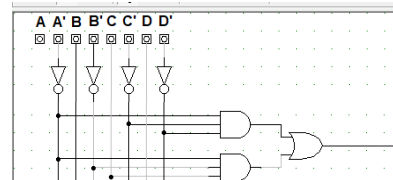
LED 5



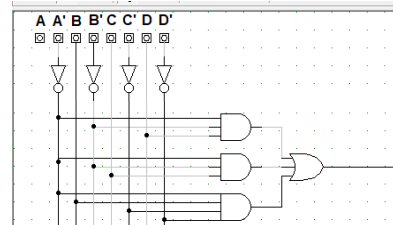
LED 6



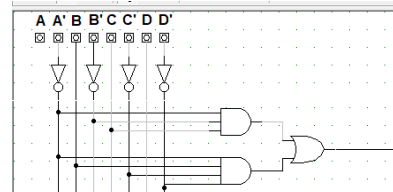
LED 7



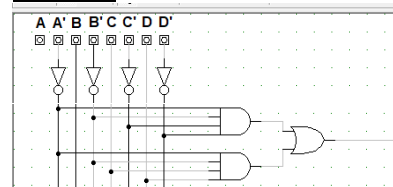
LED 8



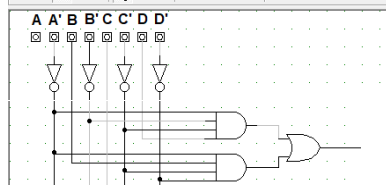
LED 9



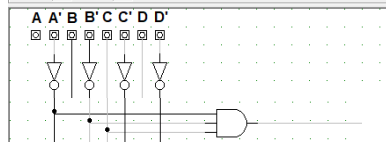
LED 10



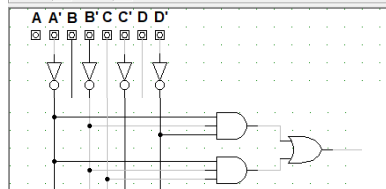
LED 11



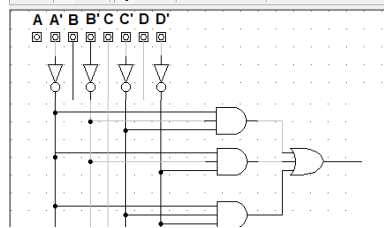
LED 12



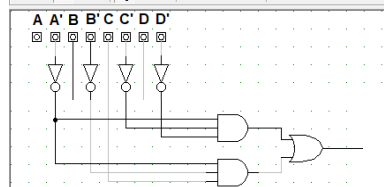
LED 13



LED 14

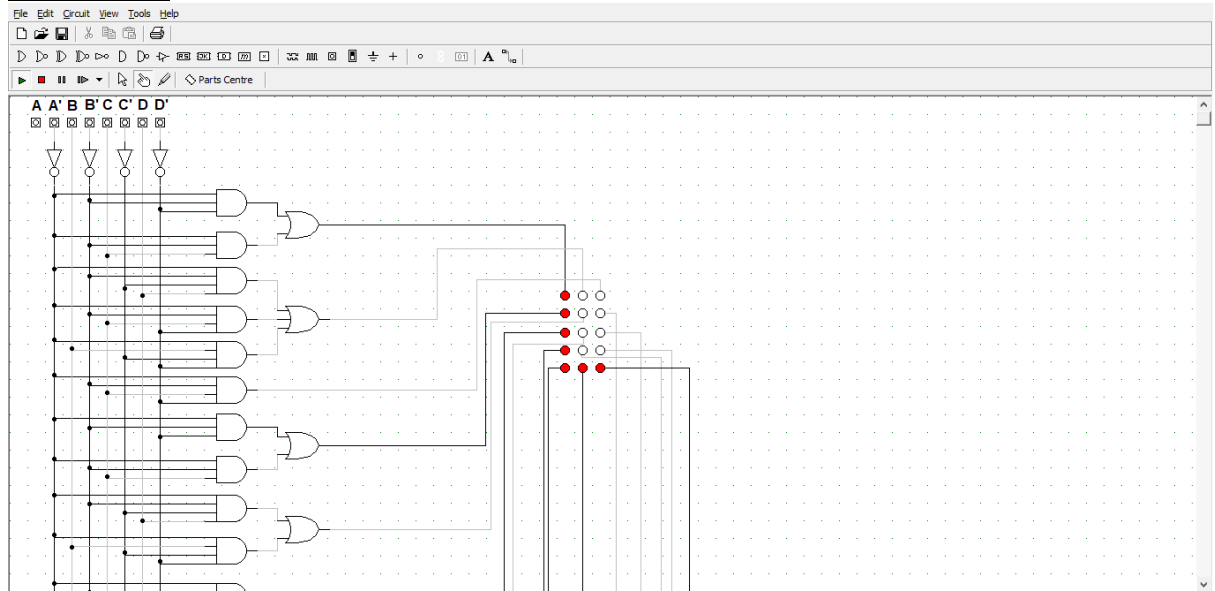


LED 15

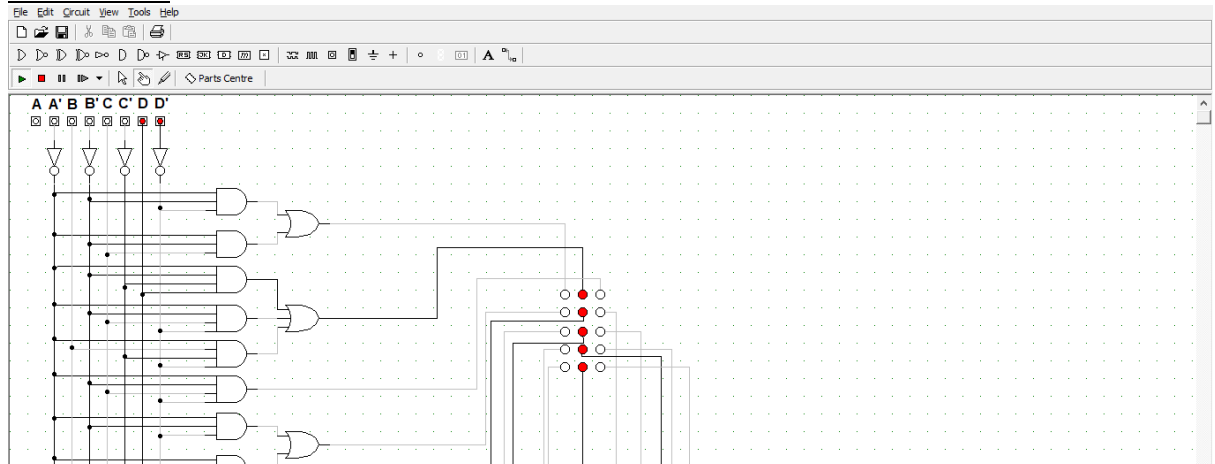


The Result Displays

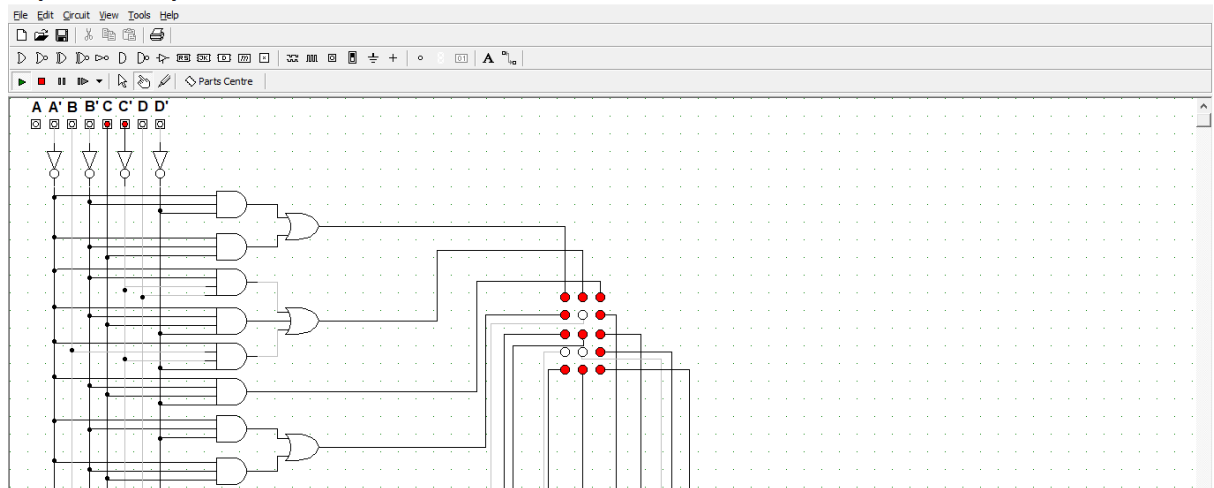
Animation 1



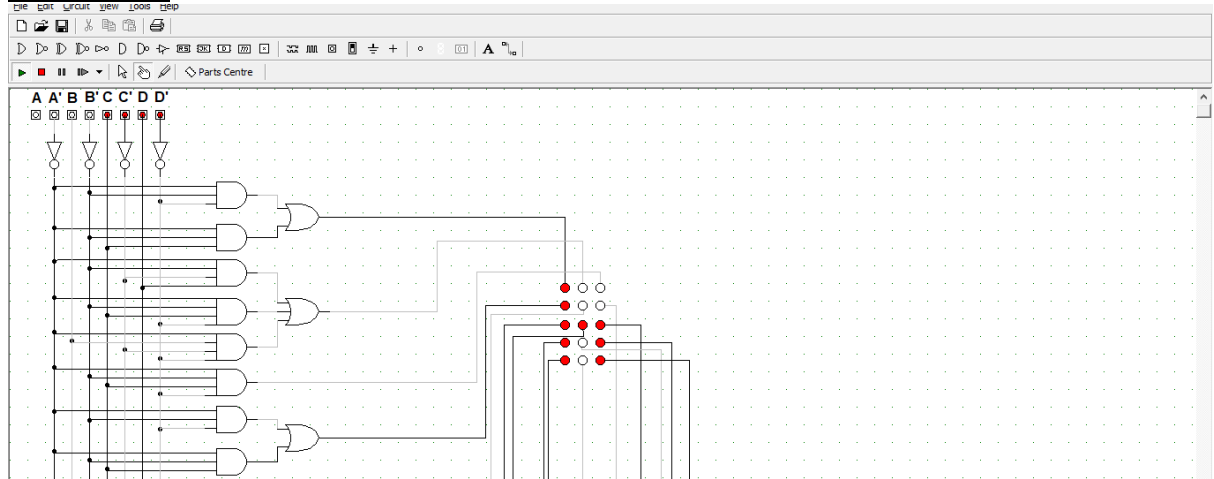
Animation 2



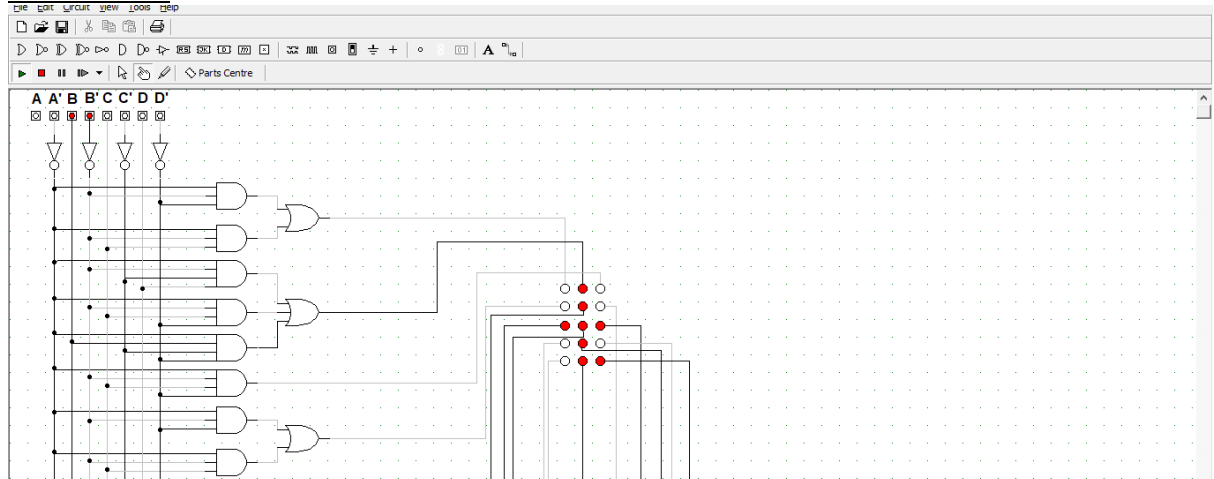
Animation 3



Animation 4



Animation 5



5. Link Github for checking Assignment Files
<https://github.com/rentsaisy/Boolean-Algebra-Midterm-Exam.git>